Gallstone Disease Risk Prediction

Gallstone Disease Risk Prediction Using Machine Learning

Data Analytics Capstone - Final Presentation

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Executive Summary

- **Objective & scope -** Build interpretable supervised models that predict gallstone status from demographic, comorbidity, body-composition, and serum biochemistry features.
- **Why it matters -** Cholelithiasis affects >700 M people worldwide and a low-cost screen using routine clinical data could surface at-risk patients before symptoms appear.
- **Data -** UCI Gallstone-1 dataset contains 319 patients (158 cases / 161 controls), 38 features, no missing values, well balanced.
- **Final results -** Random Forest CV-AUC 0.839 ± 0.048, Test AUC 0.867 and Logistic Regression Test AUC 0.881 - both well above the 0.84 Esen et al. benchmark.
- **Usage -** Reproducible pipeline informing future clinical decision support and outpatient screening protocols (not for direct patient use).

Gap Analysis

Literature establishes risk factors, but no low-cost screening tool exists.

- **Established epidemiology -** Lammert et al. (2016) and Stinton & Shaffer (2012) confirm obesity, diabetes, lipids and the four-F risk factors drive cholelithiasis.
- **Bioimpedance + biochemistry benchmark -** Esen et al. (2024) reached AUC ≈ 0.84 on the same 319-patient cohort using LR / k-NN / SVM - sets a target for this study.
- **Methodological gap -** Current clinical workflows still rely on imaging ordered after symptoms appear; no widely deployed risk score combines routinely collected features.
- **This capstone -** Re-uses routinely available demographic, body-composition and serum data to deliver transparent, interpretable LR + RF models with formal hypothesis tests.

Research Questions

Four RQs · null/alternate hypotheses - statistical test results

RQ1

Feature Distribution by Gallstone Status

Mann–Whitney U | Chi-square or Fisher (BH-FDR α=0.05)

Rejected H_0 — 24 / 31 continuous, 3 / 7 categorical features significant

RQ2

Body composition trends across BMI categories

Spearman rank correlation

Rejected H_0 — all 7 metrics monotonic; $|\rho|$ up to 0.81, $p<0.001$

RQ3

Obesity (BMI ≥ 30) vs gallstone presence

Chi-square + odds ratio with 95% CI

Failed to reject H_0 — χ^2=0.68, p=0.41, OR 1.24 [0.79–1.95]

RQ4

Can ML predict gallstone status from routine features?

Logistic Regression + Random Forest, 5-fold CV + held-out test

Rejected H_0 — CV-AUC 0.825 (LR) / 0.839 (RF); Test AUC > 0.85

All p-values adjusted with the Benjamini–Hochberg false discovery rate procedure.

Data Description and EDA

UCI Gallstone-1 · 319 patients · 38 predictors + binary target

319
patients

38
features

158 / 161
case / control

- **Balanced cohort** - no resampling needed.
- **Cases show ↑ CRP, ↑ visceral fat, ↑ total body fat**
- **Cases show ↓ vitamin D, ↓ HDL, ↓ lean mass %**
- **Hepatic fat, hyperlipidemia, gender significant after FDR**
- **Feature groups: demographics · comorbidities · anthropometry · bioimpedance · biochemistry**

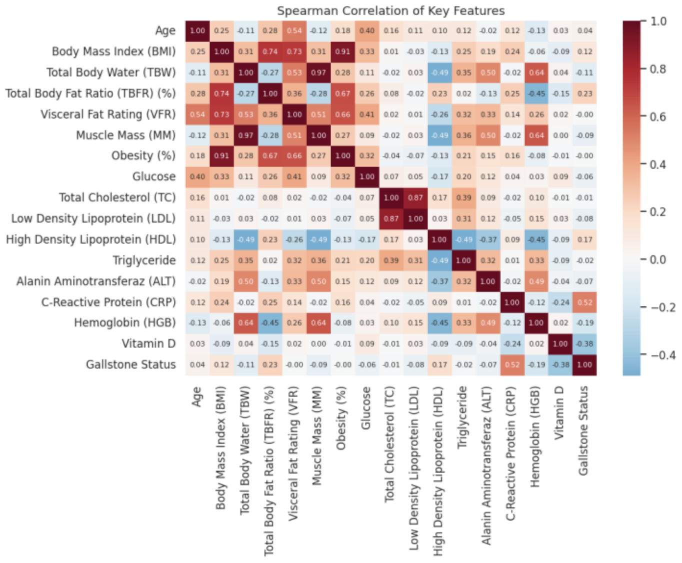
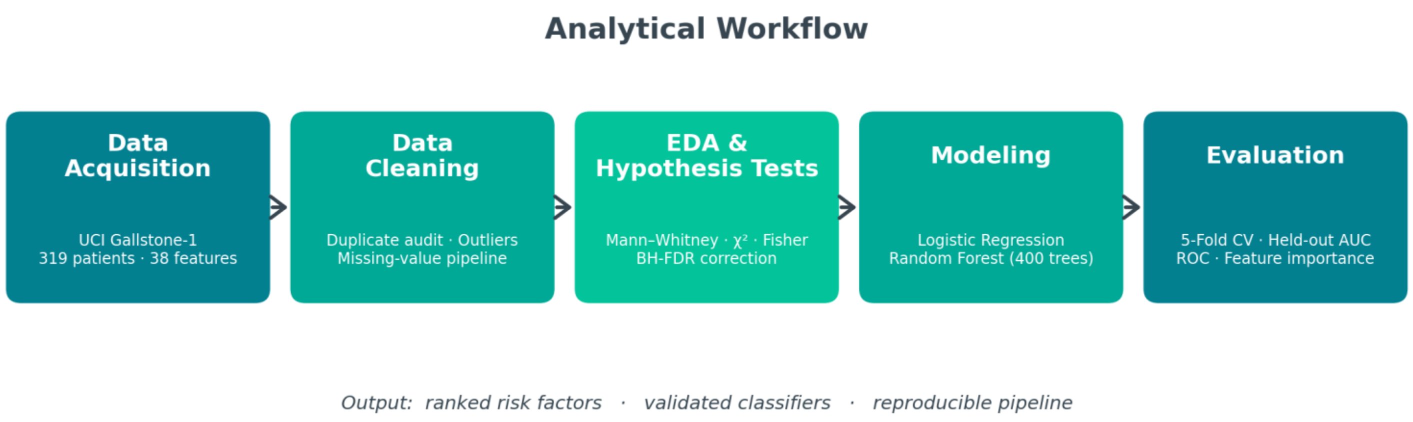


Fig 2 · Spearman correlation of key features and gallstone status

Architecture & Workflow

End-to-end reproducible pipeline implemented in scikit-learn



Stack: pandas (McKinney 2010) · scikit-learn (Pedregosa et al. 2011) · SciPy (Virtanen et al. 2020) · matplotlib / seaborn

Reproducibility: *All code, data and figures live in the public GitHub repo (github.com/prasun031087-afk/Capstone-Gallstone-Prediction).*

Model Building

Two complementary classifiers — transparent baseline + flexible ensemble

Logistic Regression · transparent baseline

- **Why -** Standardized coefficients = log-odds per σ, directly interpretable for clinicians.
- **Preprocessing -** StandardScaler on 31 continuous predictors.
- **Settings -** L2 penalty, class_weight = balanced, 80/20 stratified split.
- **Benchmark -** Mirrors Esen et al. (2024) AUC ≈ 0.84 reference.

Random Forest · non-linear ensemble

- **Why -** Captures non-linear interactions and provides Gini feature-importance ranking (Breiman, 2001).
- **Preprocessing -** No scaling required — tree splits are scale-invariant.
- **Settings -** 400 trees, class_weight = balanced, same 80/20 split.
- **Validation -** 5-fold stratified CV on training set + single held-out test.

Both models retain all 38 predictors; engineered BMI category and obesity flag are excluded to avoid target leakage.

Results

Both classifiers exceed the 0.84 literature benchmark on the held-out test set

Model	Acc.	Prec.	Recall	F1	Test AUC	CV-AUC
Logistic Regression	0.766	0.793	0.719	0.754	0.881	0.825 ± 0.055
Random Forest	0.797	0.806	0.781	0.794	0.867	0.839 ± 0.048

Held-out test n = 64 · 5-fold CV on training n = 255

- **Model Comparison -** Random Forest achieved the strongest overall classification performance demonstrating better generalization and non-linear pattern detection, while Logistic Regression achieved the highest test AUC (0.881) and provided clinically interpretable coefficient-based insights for risk assessment.
- **Top predictors -** CRP, vitamin D, ECF/TBW, AST, lean mass %, visceral fat emerged as the strongest predictors of gallstone risk across both statistical tests and machine learning models.

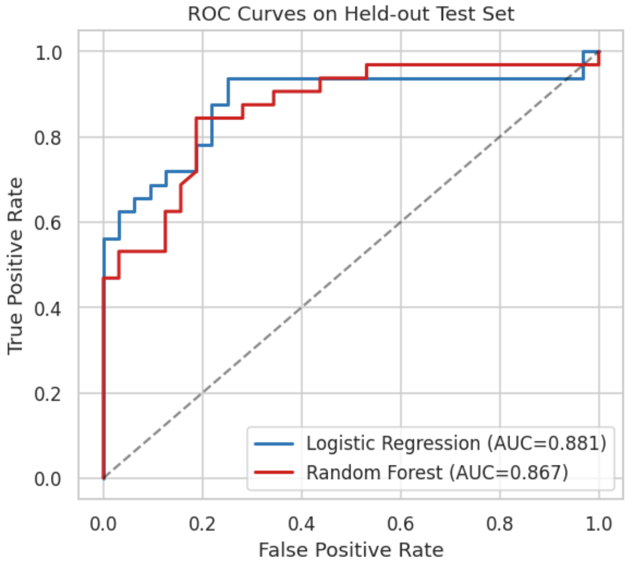


Fig 6 · ROC curves on held-out test set

Implementation & User Benefit

From notebook to a screening-ready decision support input

- **Compact screening panel -** Five routinely collected measurements - CRP, vitamin D, AST, lean mass %, visceral fat, explain most of the predictive signal.
- **Decision-support input -** Validated LR + RF pipelines can be wrapped into a triage score that surfaces at-risk patients before imaging is ordered.
- **Earlier intervention -** Surfacing high-risk patients pre-symptom may reduce emergency cholecystectomies and downstream healthcare cost.
- **Reproducible artifact -** Public GitHub repo lets clinicians, students and auditors re-run the entire analysis on the same data with one notebook.
- **Extensible foundation -** Pipeline includes SMOTE-ready imputation and class-weight handling for scaling to larger, imbalanced cohorts.

Conclusion

Findings, limitations and where this work goes next

Key findings

- Inflammation (CRP), vitamin D, AST, lean mass percentage, and visceral fat are the dominant drivers of gallstone risk in this cohort.
- Random Forest CV-AUC 0.839 ± 0.048 and Test AUC 0.867 meet the Esen et al. (2024) 0.84 benchmark.
- Detailed body-composition measures outperform a binary obesity flag (RQ3).

Limitations

- Single-center cohort of 319 patients from Ankara - external generalisability untested.
- Underweight BMI category has only 1 patient - interpret RQ2 trend with caution.

Future work

- External validation on independent multi-center cohorts.
- Calibrated probabilities for clinical use.
- Explore boosted trees and SHAP-based explanations.
- Prospective study integrating the model into outpatient triage.

Bibliography

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